

Christian Garry

Quantitative Researcher | Stochastic Control · Pricing Models · Probability & Statistics

christiangarry.southafrica@gmail.com | +44 79 3232 6827 | christiangarry.com | linkedin.com/in/christian-tt-garry

Research

Deep BSDEs for Mean-Field Market Making Durham, United Kingdom
Durham University 2026 – MSc Dissertation

- Built **models for pricing and quoting** in a competitive market-making game (Cont–Xiong) under inventory risk and mean-field competition; validated to 0.135% spread error vs an exact benchmark and scaled to $\sim 5,000$ agents.
- 20-seed multi-agent-RL (MADDPG) **trading-strategy** study: learned spreads exceed the cartel level ($t=3.22$, $p=0.0022$) — an algorithmic-collusion finding with market-surveillance relevance (FCA/PRA/SR 11-7).
- Engineered the population/competition features naive networks suppress (three fixed failure modes, 26-experiment validation); post-fix, learned competition factor varies $4\times$ and quotes shift $2\times$.
- Developed an a-posteriori error-certification theory for deep-BSDE solvers (computable estimator, two-sided guarantees, certified training-stop rule); core machine-checked in Lean 4 / Mathlib ($\sim 36k$ LOC, axiom-clean).

Silicon Carbide JFET CPU Durham, United Kingdom
MEng Dissertation (84%) Oct 2023 – Apr 2024

- Designed and simulated a 4-bit CPU in SiC JFET logic (LTspice) and built the full verification toolchain (C++ compiler, assembler, Python automation, emulator-parity + waveform-to-state checks).

Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham, United Kingdom
Durham University Sep 2025 – Exp. Sep 2026

- Tracking Distinction — average $\sim 86\%$ (95% performance/GPU, 94% ML & statistics, $\sim 89\%$ advanced Bayesian ML); coverage: Bayesian statistics, **time-series & regression** (incl. high-dim. survival regression), convex optimisation, GPU/HPC.

Mathematics Modules NetMath, Online
University of Illinois Urbana-Champaign Apr 2026 – Dec 2026

- **MATH 314** Higher Mathematics (*in progress*, 100%) & **MATH 447** Real Analysis (*upcoming*) — proof techniques, metric spaces, compactness, Riemann integration.

MEng Electronic Engineering — Upper Second-Class Honours (2:1) Durham, United Kingdom
Durham University Sep 2020 – Jun 2024

Experience

Graduate Communications Engineer Hebburn, United Kingdom
Siemens PLC Sep 2024 – Present

- Built a Python RAG platform from scratch (local LLMs, vector search) over large technical corpora — cut engineer time-to-answer by $>90\%$.
- Implemented C/C++ communication components (TCP/IP, serial) for digital-twin simulation.

Industrial Tutor Durham, United Kingdom
Durham University, Dept. of Engineering Sep 2025 – Present

- Lead weekly design tutorials, mentoring engineering teams on feasibility-driven decisions and giving structured feedback on assumptions and justification quality.

Technical Skills & Certifications

Stochastic control (BSDEs/BSDEJs, McKean–Vlasov & mean-field games, $HJB \leftrightarrow BSDE$) • probability & statistics • time-series & regression • Monte-Carlo & finite-difference numerics • optimisation • market microstructure / optimal market making • multi-agent RL (MADDPG) • Python (strong), C++, C, R • GPU/HPC • Lean 4

- **MITx 15.455x** — Mathematical Methods for Quantitative Finance (*in progress*); **Columbia** — Introduction to Financial Engineering & Risk Management.